

Week 9 Discussion: Binomial Trees (No-Arbitrage Pricing)

Main focus (50min): (i) one-step replication & risk-neutral pricing; (ii) two-step pricing via backward induction; (iii) American options and early exercise; (iv) CRR tree construction (matching volatility).

Module A) One-Step Binomial Model: Replication & Risk-Neutral Pricing

Model: $S_0 \rightarrow \begin{cases} S_0 u \\ S_0 d \end{cases}$ over T years, with $u > 1$, $d < 1$, risk-free rate r (cont. comp.).

No-arbitrage condition: $d < e^{rT} < u$.

Option payoffs: $f^u = h(S_0 u)$, $f^d = h(S_0 d)$.

Replication (static over one step):

$$\Delta_0 = \frac{f^u - f^d}{S_0 u - S_0 d}, \quad f_0 = \Delta_0 S_0 - B_0,$$

where B_0 is the amount *borrowed* (if $B_0 > 0$) or *invested* (if $B_0 < 0$) in the bank.

Risk-neutral probability and pricing:

$$p = \frac{e^{rT} - d}{u - d}, \quad f_0 = e^{-rT} (p f^u + (1 - p) f^d).$$

Interpretation: under p , the expected growth of the stock is the risk-free rate: $\mathbb{E}^Q[S_T] = S_0 e^{rT}$.

Practice A (Textbook 13.4): One-step call pricing via no-arbitrage

A stock price is currently $S_0 = \$50$. It is known that at the end of 2 months it will be either \$53 or \$48. The risk-free rate is 10% per annum with continuous compounding. What is the value of a 2-month European call option with strike $K = \$49$? Also report the hedge ratio Δ_0 .

TA Key for Practice A

Payoffs: $f^u = (53 - 49)^+ = 4$, $f^d = (48 - 49)^+ = 0$. Hence

$$\Delta_0 = \frac{4 - 0}{53 - 48} = \frac{4}{5} = 0.8.$$

Bond at maturity (use down state): $B_T = \Delta_0 \cdot 48 - f^d = 38.4$, so $B_0 = 38.4e^{-0.10 \cdot (2/12)} \approx 37.77$. Therefore

$$f_0 = \Delta_0 S_0 - B_0 = 0.8(50) - 37.77 \approx 2.23.$$

Quick Tree Construction (CRR, constant parameters)

When T is split into N equal steps of size $\Delta t = T/N$ (in years), a common choice is

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Then price by backward induction with $f = e^{-r\Delta t}(p f^u + (1 - p) f^d)$ at each node.

Module B) Two-Step Trees: Backward Induction & Dynamic Delta

Two steps: $t = 0, \Delta t, 2\Delta t$. At any node,

$$f = e^{-r\Delta t}(p f^u + (1 - p) f^d).$$

Dynamic hedging: the hedge ratio must be *rebalanced* after the first move. At a node with next-step stock prices S^u, S^d and option values f^u, f^d ,

$$\Delta = \frac{f^u - f^d}{S^u - S^d}.$$

Concept check (Textbook 13.3): Why can you *not* choose a *single* (Δ , cash) at $t = 0$ so that the position stays riskless for the whole life of a two-step option?

TA Key (Concept Check 13.3)

With two time steps, the option value after the first step depends on whether the stock moved up or down, so the required hedge ratio at $t = \Delta t$ differs across the two nodes. A *static* hedge chosen at $t = 0$ cannot simultaneously match both continuations; replication requires rebalancing (dynamic Δ).

Practice B (Textbook 13.7): Two-step European call

A stock price is currently $S_0 = \$50$. Over each of the next two 3-month periods it is expected to go up by 6% or down by 5%. The risk-free rate is 5% per annum with continuous compounding. What is the value of a 6-month European call option with strike $K = \$51$?

TA Key for Practice B

Here $\Delta t = 0.25$, $T = 0.5$, $u = 1.06$, $d = 0.95$, $r = 0.05$.

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.0125} - 0.95}{0.11} \approx \frac{1.01258 - 0.95}{0.11} \approx 0.5689.$$

Terminal stock prices: $S_{uu} = 50u^2 = 56.18$, $S_{ud} = 50ud = 50.35$, $S_{dd} = 50d^2 = 45.125$. Payoffs: $(S_{uu} - 51)^+ = 5.18$, others 0. Thus

$$C_0 = e^{-rT} p^2 \cdot 5.18 = e^{-0.025} (0.5689)^2 \cdot 5.18 \approx 1.63.$$

Module C) American Options: Early Exercise Test (Backward DP)

At maturity: $f_T = \text{payoff}$.

At each earlier node:

$$\text{continuation} = e^{-r\Delta t} (pf^u + (1-p)f^d), \quad \text{intrinsic} = \text{payoff if exercised now}.$$

$$f = \max\{\text{intrinsic}, \text{continuation}\}$$

Early exercise is optimal at a node only if intrinsic > continuation.

Practice C (Textbook 13.8): European put + parity check + American early exercise

For the situation considered in **Practice B (13.7)**, what is the value of a 6-month **European put** option with a strike price of $K = \$51$? Verify that the European call and European put prices satisfy put-call parity. If the put option were **American**, would it ever be optimal to exercise it early at any of the nodes on the tree?

TA Key for Practice C (13.8)

Same parameters as Practice B: $S_0 = 50$, $u = 1.06$, $d = 0.95$, $r = 0.05$, $\Delta t = 0.25$, $T = 0.5$.

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.0125} - 0.95}{0.11} \approx 0.5689, \quad q = 1 - p \approx 0.4311, \quad e^{-rT} = e^{-0.025}.$$

(1) **European put price.** Terminal stock prices: $S_{uu} = 56.18$, $S_{ud} = 50.35$, $S_{dd} = 45.125$. Put payoffs at T :

$$P_{uu} = (51 - 56.18)^+ = 0, \quad P_{ud} = (51 - 50.35)^+ = 0.65, \quad P_{dd} = (51 - 45.125)^+ = 5.875.$$

Hence $P_0^E = e^{-rT} (2pq \cdot 0.65 + q^2 \cdot 5.875) \approx 1.38$.

(2) **Put-call parity check.** Put-call parity (no dividends): $C_0 - P_0 = S_0 - Ke^{-rT}$. Using $C_0 \approx 1.63$ from Practice B and $P_0 \approx 1.38$,

$$C_0 - P_0 \approx 0.25, \quad S_0 - Ke^{-rT} = 50 - 51e^{-0.025} \approx 0.259,$$

which matches up to rounding.

(3) **If the put were American: early exercise?** Backward induction one step earlier ($t = \Delta t$):

$$S_u = 50u = 53, \quad S_d = 50d = 47.5, \quad e^{-r\Delta t} = e^{-0.0125}.$$

At the **up node** ($S = 53$): intrinsic = $(51 - 53)^+ = 0$. Continuation $\text{cont}_u = e^{-r\Delta t} (p \cdot 0 + q \cdot 0.65) = e^{-0.0125} (0.65q) \approx 0.277$, so **do not** exercise.

At the **down node** ($S = 47.5$): intrinsic = $51 - 47.5 = 3.5$. Continuation $\text{cont}_d = e^{-r\Delta t} (p \cdot 0.65 + q \cdot 5.875) \approx 2.87$, so **exercise early** at this node.

Therefore

$$P_0^{Am} = e^{-r\Delta t} (p \cdot P_u^{Am} + q \cdot P_d^{Am}) = e^{-0.0125} (p \cdot 0.277 + q \cdot 3.5) \approx 1.65.$$